Four-zero tail-palindromic primitive polynomials: coefficient geometry, Dickson reduction, and computational evidence

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Abstract

Let $n = 2k$. A polynomial $f(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0$ over $\mathbb{F}_2$ is called a *four-zero tail-palindromic candidate* if the tail vector $(c_{n-1}, \ldots, c_0)$ is a palindrome with exactly four zero entries. We establish three layers of structure for these candidates. First, we prove the self-paired identity $H_{\text{even}} = Q_k \oplus V_k$, determine the number of occupied K-cosets, construct a half-space functional, and give a complete Fourier description of the four-zero coset multiplicity. Second, we prove a uniform Dickson-factor reduction valid for every even degree and specialize it to the $n = 2p$ subfamily. Third, we report complete counts up to $n = 200$, first solutions up to $n = 300$, and a negative finite-pair association signal in the computed range for the $n = 2p$ family. Auxiliary combinatorial and arithmetic descriptions are collected in an appendix. The main conjecture remains open.

1 Introduction

Let $\mathbb{F}_2$ be the field with two elements. For an even integer $n = 2k$, write

$$f(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0, \qquad c_i \in \mathbb{F}_2.$$

The *tail vector* is $(c_{n-1}, \ldots, c_0)$. We say that f is a *four-zero tail-palindromic candidate* if

$$(c_{n-1}, \ldots, c_0) = (c_0, \ldots, c_{n-1})$$

and the tail vector has exactly four zero entries. Equivalently, for some

$$1 \le i < j \le k - 1,$$

one has

$$f_{n,i,j}(x) = S_n(x) + x^i + x^{n-1-i} + x^j + x^{n-1-j}, \qquad S_n(x) = 1 + x + \cdots + x^n. \tag{1}$$

Remark 1.1. Set $g(x) = xf(x) + 1$. The tail-palindromic condition for f is equivalent to g being self-reciprocal of degree $n + 1$:

$$x^{n+1}g(1/x) = g(x).$$

Thus the coefficient pattern studied here is a reversal-symmetric slice of polynomial space. However, primitivity of f does not imply irreducibility or primitivity of g, so we use this identification only as a structural and literature bridge, not as a reduction.

Conjecture 1.2 (C_4)**.** For every even $n \geq 6$, there exists a primitive polynomial over $\mathbb{F}_2$ of the form (1).

Conjecture 1.2 lies in the general area of primitive polynomials with prescribed coefficients. Hansen–Mullen type results prescribe one coefficient; results of Cohen and Han prescribe a small number of coefficients; Fan's theorem prescribes the last half of the coefficients when q is sufficiently large [4, 5, 6, 7, 8]. These results do not directly cover the fixed-field, growing-degree situation with four prescribed zero coefficients and palindromic symmetry. The purpose of this paper is to record and organize the structural reductions that the problem admits, together with computational evidence. Standard finite-field background may be found in [10]; the existence of primitive elements with prescribed trace in the even-characteristic setting is classical [1, 2, 3]; a related cycle-structure construction appears in [9].

The main structural results are Theorems 3.1, 4.1, and 5.1, Proposition 4.3, and the uniform Dickson-factor reduction of Section 6; auxiliary combinatorial and arithmetic descriptions are given in Appendix A.

2 Preliminaries

Let $V = \mathbb{F}_2^{2k}$. We write a vector $W \in V$ in two halves

$$W = AB, \qquad A, B \in \mathbb{F}_2^k.$$

Let $\mathrm{rev} : \mathbb{F}_2^k \to \mathbb{F}_2^k$ denote reversal of coordinates. Define

$$S(AB) = BA, \qquad R(AB) = \mathrm{rev}(B)\,\mathrm{rev}(A), \qquad P = SR.$$

The common fixed set of S, R, P will be denoted by K; explicitly,

$$K = \{CC : C = \mathrm{rev}(C)\}.$$

Let

$$f(C) = C + \mathrm{rev}(C).$$

Let $J = (1, \ldots, 1) \in \mathbb{F}_2^k$, and let $\langle \cdot, \cdot \rangle$ denote the standard dot product on $\mathbb{F}_2^k$. The self-paired set is

$$H = \{W = AB \in V : f(A) = f(B)\}.$$

We use zero-based indices for the standard unit vectors: $e_r = (0, \ldots, 0, 1, 0, \ldots, 0)$ with the nonzero entry in position r, $0 \leq r \leq k-1$.

The following elementary facts are used repeatedly.

Lemma 2.1. *The set K is a linear subspace of V of dimension $\lceil k/2 \rceil$; in particular*

$$|K| = 2^{\lceil k/2 \rceil}.$$

Moreover $K \subseteq H$.

Proof. A vector $C = (c_0, \ldots, c_{k-1})$ is fixed by reversal precisely when $c_i = c_{k-1-i}$. Thus the free coordinates are the first $\lceil k/2 \rceil$ entries. For $C = \mathrm{rev}(C)$, the word CC satisfies $f(C) = C + \mathrm{rev}(C) = 0$, so its two halves have the same image under f; hence $CC \in H$. □

Lemma 2.2. *The set H is a linear subspace of dimension $k + \lceil k/2 \rceil$. Hence*

$$|H| = 2^{k+\lceil k/2 \rceil}.$$

Proof. The condition $f(A) = f(B)$ is linear. The map $(A, B) \mapsto f(A) + f(B)$ has image $\operatorname{im} f$, of dimension $\lfloor k/2 \rfloor$. Therefore

$$\dim H = 2k - \lfloor k/2 \rfloor = k + \lceil k/2 \rceil.$$

□

3 The even self-paired target

Let

$$Q_k = \{AA : A \in \mathbb{F}_2^k\}, \qquad V_k = \operatorname{im} f = \{C + \operatorname{rev}(C) : C \in \mathbb{F}_2^k\}.$$

We identify V_k with the subspace $\{(v, 0) : v \in V_k\} \subseteq V$.

Theorem 3.1. *One has*

$$\boxed{H_{\text{even}} = Q_k \oplus V_k}$$

where H_{even} is the subset of H consisting of vectors of even Hamming weight. In particular

$$|H_{\text{even}}| = |Q_k|\,|V_k| = 2^{k+\lfloor k/2 \rfloor} = 2^{2k-\lceil k/2 \rceil}.$$

Proof. Let $W = (C + v, C)$ with $C \in \mathbb{F}_2^k$ and $v \in V_k$. Since $V_k = \operatorname{im} f \subseteq \ker f$, we have $f(v) = 0$, hence

$$f(C + v) + f(C) = f(v) = 0.$$

Thus $W \in H$. Also

$$\operatorname{wt}(C + v) + \operatorname{wt}(C) \equiv \operatorname{wt}(v) \equiv 0 \pmod 2,$$

because $v = f(D) = D + \operatorname{rev}(D)$ has even weight. Therefore $Q_k \oplus V_k \subseteq H_{\text{even}}$.

It remains to compare sizes. Let $W = AB \in H$ and put $z = A + B$. Then $z \in \ker f$. The parity of $\operatorname{wt}(W)$ equals the parity of $\operatorname{wt}(z)$, which is the value of the linear functional $\ell(z) = \langle J, z \rangle$ on $\ker f$.

If k is even, then $\dim \ker f = \dim \operatorname{im} f = k/2$. Moreover every vector in $\operatorname{im} f$ has even weight, so $\operatorname{im} f \subseteq \ker \ell$. Since the dimensions are equal, $\ker f = \operatorname{im} f$ and ℓ vanishes on $\ker f$. Hence all elements of H have even weight.

If k is odd, then ℓ is nonzero on $\ker f$, because $J \in \ker f$ and $\ell(J) = \operatorname{wt}(J) = k \equiv 1 \pmod 2$. Hence exactly half of the elements of $\ker f$ lie in $\ker \ell$. The map $W = AB \mapsto A + B$ from H to $\ker f$ is surjective: for any $z \in \ker f$, the word $z0$ lies in H and maps to z. Since H_{even} is the preimage of $\ker \ell$ under this map, it is a hyperplane in H; consequently $|H_{\text{even}}| = \frac{1}{2}|H|$.

Combining these cases with Lemma 2.2 gives

$$|H_{\text{even}}| = \begin{cases} 2^{k+k/2}, & k \text{ even}, \\ \frac{1}{2} 2^{k+(k+1)/2}, & k \text{ odd}, \end{cases}$$

which in both cases equals $2^{k+\lfloor k/2 \rfloor} = 2^{2k-\lceil k/2 \rceil}$.

Finally, $|Q_k| = 2^k$ and $|V_k| = 2^{\lfloor k/2 \rfloor}$, and the sum is direct because the two halves of an element of Q_k are equal, while a nonzero element of V_k (identified with $\{(v, 0)\}$) has second half 0. Thus

$$|Q_k \oplus V_k| = 2^{k+\lfloor k/2 \rfloor} = |H_{\text{even}}|.$$

The already established inclusion $Q_k \oplus V_k \subseteq H_{\text{even}}$ is therefore an equality. □

Corollary 3.2. *Every four-zero tail-palindromic candidate belongs to* H_{even}.

Proof. The tail vector of a four-zero candidate is a palindrome and has exactly four zero entries, hence its Hamming weight is even. A palindrome satisfies $B = \text{rev}(A)$, so $f(A) = A + B = f(B)$; hence it lies in H. □

Remark 3.3. The four-zero candidates do not exhaust H_{even}. They form a thin, polynomially-sized subfamily, whereas H_{even} is exponentially large.

4 The K-coset invariant and its count

Write

$$\mathcal{G} = H_{\text{even}}/K$$

for the quotient of the self-paired even-weight target by the common fixed-point kernel.

For a four-zero candidate, write $A = J + e_i + e_j$, where $J = (1, \ldots, 1)$ and $1 \le i < j \le k-1$. The candidate is $W = A\,\text{rev}(A)$. Its K-coset is determined by

$$g(A) = A + \text{rev}(A) \in V_k.$$

Since J is palindromic,

$$g(A) = g(e_i + e_j).$$

Let $D(k)$ be the number of distinct K-cosets occupied by four-zero candidates.

Theorem 4.1. *For* $k \ge 4$,

$$\boxed{D(k) = 1 + \binom{\lfloor (k+1)/2 \rfloor}{2}.}$$

For $k = 3$, $D(3) = 1$. *Consequently, for* $k \ge 4$,

$$D(k+2) - D(k) = \left\lfloor \frac{k+1}{2} \right\rfloor.$$

Proof. Write $g(e_i + e_j) = v_i + v_j$, where

$$v_r = e_r + e_{k-1-r}.$$

If k is odd and $r = (k-1)/2$, then $v_r = 0$; otherwise v_r is the pair-vector of the symmetric pair $\{r, k-1-r\}$. Thus, after identifying v_r with v_{k-1-r}, every possible value of $g(e_i + e_j)$ is a sum of at most two distinct pair-vectors. If $k = 2m$ is even, there are m pair-vectors and no fixed coordinate. The possible values are 0 and sums of two distinct pair-vectors, whence

$$D(2m) = 1 + \binom{m}{2}.$$

If $k = 2m + 1$ is odd, there are m nonzero pair-vectors and one fixed middle coordinate, giving $v_{(k-1)/2} = 0$. The possible values are 0, a single nonzero pair-vector, and a sum of two distinct nonzero pair-vectors; hence

$$D(2m+1) = 1 + m + \binom{m}{2} = 1 + \binom{m+1}{2}.$$

The two formulas combine into the stated expression. The case $k = 3$ is exceptional because the unique candidate gives $g(e_1 + e_2) \neq 0$, so the value 0 is not attained and $D(3) = 1$ while the general formula would give 2. The increment formula follows by subtraction. □

For $1 \le i \le k-1$, set

$$W_i = (J + e_i)(J + e_{k-1-i}), \qquad u_i = \overline{W}_i \in \mathcal{G}.$$

Lemma 4.2. *With the notation just defined, write $r = \lfloor k/2 \rfloor$. As a multiset,*

$$\{u_1, \dots, u_{k-1}\} = \begin{cases} \{z_1, z_1, \dots, z_{r-1}, z_{r-1}, z_r\}, & k = 2r, \\ \{z_1, z_1, \dots, z_{r-1}, z_{r-1}, 0, z_r\}, & k = 2r+1, \end{cases}$$

where $z_1, \dots, z_r$ are linearly independent in $\mathcal{G}$.

Proof. First note that each W_i lies in H_{even}: its two halves are interchanged by reversal, so their f-images agree, and its Hamming weight is $2(k-1)$. Now for $1 \le i \le r-1$, the two elements W_i and W_{k-1-i} differ by

$$(e_i + e_{k-1-i},\, e_i + e_{k-1-i}),$$

whose common half is palindromic. Hence their difference lies in K, so $u_i = u_{k-1-i}$.

If $k = 2r+1$, then

$$W_r = (J + e_r)(J + e_r) \in K,$$

so $u_r = 0$.

It remains to see that the remaining elements are independent. Under the decomposition $H_{\text{even}} = Q_k \oplus V_k$, the image of u_i in the quotient $(Q_k/K) \oplus V_k$ has V_k-component $e_i + e_{k-1-i}$. For $i = 1, \dots, r-1$, these are the pair-vectors $e_i + e_{k-1-i}$, and the remaining element u_{k-1} has V_k-component $e_0 + e_{k-1}$. These r pair-vectors form a basis of V_k, so the corresponding elements of $\mathcal{G}$ are linearly independent. □

Proposition 4.3 (Fourier spectrum)**.** *Recall that $\mathcal{G} = H_{\text{even}}/K$. For $1 \le i \le k-1$, recall that*

$$W_i = (J + e_i)(J + e_{k-1-i}), \qquad u_i = \bar{W}_i \in \mathcal{G}.$$

Then $W_i \in H_{\text{even}}$, and for every $1 \le i < j \le k-1$,

$$\bar{W}_{i,j} = u_i + u_j. \tag{2}$$

Consequently, for every character $\chi \in \widehat{\mathcal{G}}$,

$$\widehat{\mu}(\chi) = \frac{1}{2}\left(S_\chi^2 - (k-1)\right), \qquad S_\chi = \sum_{i=1}^{k-1} \chi(u_i). \tag{3}$$

Fourier inversion therefore gives

$$\mu(w) = \frac{1}{|\mathcal{G}|} \sum_{\chi \in \widehat{\mathcal{G}}} \frac{S_\chi^2 - (k-1)}{2}\, \chi(w). \tag{4}$$

Proof. For $A_i = J + e_i$ one has $\operatorname{rev}(A_i) = J + e_{k-1-i}$, so $W_i = A_i \operatorname{rev}(A_i) \in H$. Its Hamming weight is $2(k-1)$, hence even, and thus $W_i \in H_{\text{even}}$.

For $1 \le i < j \le k-1$, the four-zero candidate has the form

$$W_{i,j} = (J + e_i + e_j)(J + e_{k-1-i} + e_{k-1-j}).$$

Therefore

$$W_{i,j} + W_i + W_j = (J, J).$$

Since J is palindromic, $(J, J) \in K$. Passing to H_{even}/K gives (2).

Now let $\chi \in \widehat{\mathcal{G}}$. Using additivity of χ,

$$\widehat{\mu}(\chi) = \sum_{i<j} \overline{\chi(u_i + u_j)} = \sum_{i<j} \chi(u_i)\chi(u_j),$$

because $\chi(u_i) = \pm 1$. Hence

$$\widehat{\mu}(\chi) = \frac{1}{2}\left(\left(\sum_{i=1}^{k-1} \chi(u_i)\right)^2 - (k-1)\right),$$

which is (3). Equation (4) is the Fourier inversion formula on the finite abelian group $\mathcal{G}$. □

Corollary 4.4 (Exact multiplicity distribution)**.** *Fix the independent elements $z_1, \ldots, z_r$ from Lemma 4.2. For*

$$w = \sum_{j=1}^{r} \epsilon_j z_j, \qquad \epsilon = (\epsilon_1, \ldots, \epsilon_r) \in \mathbb{F}_2^r,$$

the multiplicity $\mu(w)$ is as follows.

If $k = 2r$, then

$$\mu(w) = \begin{cases} r-1, & \epsilon = 0, \\ 2, & \epsilon = \varepsilon^{(j)} + \varepsilon^{(r)}, \quad 1 \le j < r, \\ 4, & \epsilon = \varepsilon^{(j)} + \varepsilon^{(\ell)}, \quad 1 \le j < \ell < r, \\ 0, & \textit{otherwise.} \end{cases}$$

If $k = 2r + 1$, then

$$\mu(w) = \begin{cases} r-1, & \epsilon = 0, \\ 2, & \epsilon = \varepsilon^{(j)}, \quad 1 \le j < r, \\ 1, & \epsilon = \varepsilon^{(r)}, \\ 2, & \epsilon = \varepsilon^{(j)} + \varepsilon^{(r)}, \quad 1 \le j < r, \\ 4, & \epsilon = \varepsilon^{(j)} + \varepsilon^{(\ell)}, \quad 1 \le j < \ell < r, \\ 0, & \textit{otherwise.} \end{cases}$$

Proof. By Lemma 4.2, the multiset of u's consists of two copies of each z_j for $1 \le j < r$, together with the additional element z_r. If k is odd, there is also one zero element. A pair $u_i + u_j$ can therefore have only the following forms: two copies of the same z_j give 0; one copy of z_j paired with z_r gives $z_j + z_r$; one copy of z_j paired with one copy of z_ℓ gives $z_j + z_\ell$; and, when k is odd, the zero element paired with z_j gives z_j. Counting the available copies gives the stated multiplicities. □

Corollary 4.5 (All moments)**.** *For every integer $t \ge 1$,*

$$M_t := \sum_{w \in \mathcal{G}} \mu(w)^t = \begin{cases} (r-1)^t + 2^t(r-1) + 4^t\binom{r-1}{2}, & k = 2r, \\ (r-1)^t + 2^t(2r-2) + 1 + 4^t\binom{r-1}{2}, & k = 2r+1. \end{cases}$$

In particular, the case $t = 2$ gives the second moment below.

Proof. This follows immediately from the exact multiplicity distribution in Corollary 4.4. □

Corollary 4.6 (Closed-form second moment)**.** *Write $k = 2r$ or $k = 2r + 1$, and put*

$$C_2 = \sum_{w \in \mathcal{G}} \mu(w)^2.$$

Then

$$C_2 = \begin{cases} (r-1)(9r-13), & k = 2r, \\ 9r^2 - 18r + 10, & k = 2r+1. \end{cases} \tag{5}$$

Proof. Apply the distribution in Corollary 4.4. If $k = 2r$, then

$$C_2 = (r-1)^2 + 4(r-1) + 16\binom{r-1}{2} = (r-1)(9r-13).$$

If $k = 2r + 1$, then

$$C_2 = (r-1)^2 + 8(r-1) + 1 + 16\binom{r-1}{2} = 9r^2 - 18r + 10.$$

□

Remark 4.7 (Ambient check-class quotient)*.* Since $K \subseteq H_{\text{even}}$, the quotient studied above is a subgroup of the ambient quotient $\mathbb{F}_2^{2k}/K$. Their dimensions are

$$\dim_{\mathbb{F}_2} \frac{\mathbb{F}_2^{2k}}{K} = 2k - \left\lceil \frac{k}{2} \right\rceil, \qquad \dim_{\mathbb{F}_2} \frac{H_{\text{even}}}{K} = 2 \left\lfloor \frac{k}{2} \right\rfloor.$$

Consequently,

$$\left[\frac{\mathbb{F}_2^{2k}}{K} : \frac{H_{\text{even}}}{K} \right] = 2^{\lceil k/2 \rceil}.$$

For $k = 3$, this is the familiar 16-class ambient quotient containing the four-element subgroup H_{even}/K.

5 A half-space functional

Let $W = AB \in H$. Define

$$s(W) = \text{the first coordinate of } \operatorname{rev}(A) + B \in \mathbb{F}_2^k.$$

Theorem 5.1. *The map s is well defined on H/K, is a nonzero linear functional on H/K, and satisfies*

$$s(W) = 0$$

for every four-zero candidate. For the generalized alternating direction $E = 0101\cdots01$ (a full word of length $2k$), one has $s(E) = 1$.

Proof. If W is replaced by $W + CC$ with $C = \operatorname{rev}(C)$, then A and B are each replaced by $A + C$ and $B + C$. Since $\operatorname{rev}(C) = C$, the expression $\operatorname{rev}(A) + B$ is unchanged. This proves well-definedness. For a four-zero candidate one has $B = \operatorname{rev}(A)$, so $\operatorname{rev}(A) + B = 0$ and $s(W) = 0$. For the alternation vector E, the first coordinate of $\operatorname{rev}(E_A) + E_B$ is 1, so $s(E) = 1$. □

Corollary 5.2. *All four-zero candidates lie in the half-space $s = 0$, while the generalized alternating direction lies in the opposite half-space $s = 1$.*

6 Uniform Dickson reduction for every even degree

We first record the arithmetic reduction in its natural generality. The prime-degree specialization is treated separately in Section 7.

Let

$$n = 2k, \qquad q = 2^k, \qquad \mathbb{K} = \mathbb{F}_q, \qquad \mathbb{L} = \mathbb{F}_{q^2}.$$

Since $q^2 - 1 = (q-1)(q+1)$ and $\gcd(q-1, q+1) = 1$, there is a direct decomposition

$$\mathbb{L}^* \cong \mathbb{K}^* \times \mathbb{H}_B, \qquad \mathbb{H}_B = \{x \in \mathbb{L}^* : x^{q+1} = 1\}.$$

We use the Dickson polynomials $D_r(t)$, defined by

$$D_0(t) = 0, \qquad D_1(t) = t, \qquad D_{r+1}(t) = tD_r(t) + D_{r-1}(t),$$

so that

$$D_r(u + u^{-1}) = u^r + u^{-r}.$$

Proposition 6.1 (Uniform τ-reduction)**.** *Let $\beta \in \mathbb{L}^*$ be primitive and write uniquely*

$$\beta = \nu\gamma, \qquad \nu \in \mathbb{K}^*, \quad \gamma \in \mathbb{H}_B.$$

Then

$$\operatorname{ord}(\nu) = q - 1, \qquad \operatorname{ord}(\gamma) = q + 1.$$

Put $\tau = \gamma + \gamma^{-1} \in \mathbb{K}$. Then

$$D_q(\tau) = \tau, \qquad D_{q+1}(\tau) = 0,$$

and the minimal polynomial of β over $\mathbb{F}_2$ is

$$F_{\gamma,\nu}(x) = \prod_{i=0}^{k-1} \left(x^2 + \nu^{2^i} D_{2^i}(\tau)x + \nu^{2^{i+1}}\right). \tag{6}$$

Moreover, if s_r denotes the r-th power sum of the roots of $F_{\gamma,\nu}$, then

$$s_r = \operatorname{Tr}_{\mathbb{K}/\mathbb{F}_2}\left(\nu^r D_r(\tau)\right). \tag{7}$$

Proof. The order statements follow from the direct decomposition and the fact that β generates $\mathbb{L}^*$. The Dickson identities are

$$D_q(\gamma + \gamma^{-1}) = \gamma^q + \gamma^{-q} = \gamma^{-1} + \gamma = \tau,$$

and

$$D_{q+1}(\gamma + \gamma^{-1}) = \gamma^{q+1} + \gamma^{-(q+1)} = 1 + 1 = 0,$$

where $\gamma^q = \gamma^{-1}$.

The Frobenius conjugates of β are $\nu^{2^i}\gamma^{2^i}$, $0 \le i \le 2k-1$. Since $\nu^q = \nu$ and $\gamma^q = \gamma^{-1}$, these roots pair as

$$\nu^{2^i}\gamma^{2^i}, \qquad \nu^{2^i}\gamma^{-2^i}, \qquad 0 \le i \le k-1.$$

Their sum is $\nu^{2^i} D_{2^i}(\tau)$ and their product is $\nu^{2^{i+1}}$, giving (6). The power-sum identity (7) follows by summing the same paired roots and applying the definition of D_r. □

Corollary 6.2. *For every even $n = 2k$, the four-zero condition is equivalent to the system*

$$\begin{cases} c_r(\nu,\tau) = 0, & r \in \{i, j, n-1-i, n-1-j\}, \\ c_r(\nu,\tau) = 1, & r \notin \{i, j, n-1-i, n-1-j\}, \\ \nu \in \mathbb{K}^*, & \operatorname{ord}(\nu) = q-1, \\ \gamma \in \mathbb{H}_B, & \operatorname{ord}(\gamma) = q+1, \quad \tau = \gamma + \gamma^{-1}, \end{cases} \tag{8}$$

where the $c_r(\nu,\tau)$ are the coefficients of $F_{\gamma,\nu}$.

For fixed $i < j$, write

$$c_r(\nu, T) = \nu^{a_r} \tilde{c}_r(\nu, T), \qquad \tilde{c}_r(0, T) \neq 0, \qquad r \in \{2k-1-i, 2k-1-j\},$$

and define the reduced resultants

$$R_{i,j}^{(k)}(T) = \operatorname{Res}_\nu\big(\tilde{c}_{2k-1-i}(\nu, T), \tilde{c}_{2k-1-j}(\nu, T)\big), \qquad M_k(T) = \prod_{1 \le i < j \le k-1} R_{i,j}^{(k)}(T).$$

Set

$$\mathcal{S}_k(T) = \gcd\big(G_k(T), M_k(T)\big), \qquad G_k(T) = \gcd(D_q(T) + T, D_{q+1}(T)),$$

with the convention that $M_k = 0$ if some resultant is identically zero. Then every solution of the general $n = 2k$ system (8) forces the minimal polynomial of τ to divide $\mathcal{S}_k(T)$. This gives a necessary filter for every even degree. The Frobenius orbit of τ has length dividing k; in the prime half-degree case $k = p$ treated in Section 7, it has length exactly p. An auxiliary gcd identity for the symmetric coefficient weights is recorded in Appendix A; it is not used in this reduction.

7 The subfamily $n = 2p$

We now specialize the general reduction of Section 6 to the prime half-degree $k = p$. Thus

$$q = 2^p, \qquad \mathbb{K} = \mathbb{F}_q, \qquad \mathbb{L} = \mathbb{F}_{q^2}, \qquad \mathbb{H}_B = \{x \in \mathbb{L}^* : x^{q+1} = 1\}.$$

If $\beta = \nu\gamma$ is primitive, Proposition 6.1 gives

$$\operatorname{ord}(\nu) = q - 1 = 2^p - 1, \qquad \operatorname{ord}(\gamma) = q + 1 = 2^p + 1.$$

Set $\tau = \gamma + \gamma^{-1} \in \mathbb{K}$. The Frobenius conjugates satisfy

$$\tau_r = \gamma^{2^r} + \gamma^{-2^r} = \tau^{2^r},$$

so a four-zero primitive polynomial determines the Frobenius orbit of τ, equivalently the minimal polynomial of τ over $\mathbb{F}_2$.

The uniform Dickson identities become

$$D_{2^p}(\tau) = \tau, \qquad D_{2^p+1}(\tau) = 0.$$

Therefore the minimal polynomial $m_\tau(T)$ of τ divides

$$G_p(T) = \gcd\big(D_{2^p}(T) + T,\ D_{2^p+1}(T)\big). \tag{9}$$

The uniform factorization specializes to

$$F_{\gamma,\nu}(x) = N_{\mathbb{K}/\mathbb{F}_2}\big(x^2 + \nu\tau x + \nu^2\big),$$

whose roots are

$$\nu^{2^i}\gamma^{2^i}, \qquad \nu^{2^i}\gamma^{-2^i}, \qquad i = 0, \ldots, p-1.$$

Its power sums are

$$s_r = \mathrm{Tr}_{\mathbb{K}/\mathbb{F}_2}\left(\nu^r D_r(\tau)\right). \tag{10}$$

Expanding the product (6) directly, every coefficient $c_r(\nu, \tau)$ of $F_{\gamma,\nu}$ is an explicit polynomial in ν, τ.

Let the four-zero tail-palindromic condition for a pair $i < j$ be the full system

$$\begin{cases} c_r(\nu,\tau) = 0, & r \in \{i, j, 2p-1-i, 2p-1-j\}, \\ c_r(\nu,\tau) = 1, & r \notin \{i, j, 2p-1-i, 2p-1-j\}, \\ \nu \in \mathbb{K}^*, \quad \mathrm{ord}(\nu) = 2^p - 1, & \\ \gamma \in \mathbb{H}_B, \quad \mathrm{ord}(\gamma) = 2^p + 1, \quad \tau = \gamma + \gamma^{-1}. & \end{cases} \tag{11}$$

Equation (11) is the exact coefficient-and-order formulation of the $n = 2p$ subfamily.

For fixed $i < j$, write

$$c_r(\nu, T) = \nu^{a_r}\tilde{c}_r(\nu, T), \qquad \tilde{c}_r(0, T) \neq 0,$$

and define

$$R_{i,j}(T) = \mathrm{Res}_\nu\left(\tilde{c}_{2p-1-i}(\nu, T),\, \tilde{c}_{2p-1-j}(\nu, T)\right).$$

These two equations are necessary conditions for the full system (11); hence the resulting filter is necessary but not sufficient. Here ν is regarded as a formal variable; the order conditions on ν and γ are not encoded. Set

$$M_p(T) = \prod_{1 \le i < j \le p-1} R_{i,j}(T).$$

Then

$$\mathcal{S}_p(T) = \gcd(G_p(T), M_p(T))$$

is a *necessary filter*: if the full system (11) has a solution for a given τ-orbit, then the corresponding minimal polynomial divides $\mathcal{S}_p(T)$. The reverse implication is not asserted: M_p does not encode all coefficient equalities in (11), nor does it encode the order conditions on ν and γ. Thus

$$\mathcal{S}_p(T) \neq 1$$

is a necessary condition for the existence of a four-zero candidate in the $n = 2p$ family, but not by itself a sufficient one. A complete resultant system would have to eliminate all equations of (11), not only the two high coefficient equations.

8 Computational evidence

8.1 Exact counts

The four-zero candidates were enumerated for all even $6 \le n \le 200$. The total number of primitive candidates is 2663, and every layer has $N_4(n) > 0$. First-solution searches extend to $n \le 300$. Table 1 gives a representative sample.

n	candidates $\binom{n/2-1}{2}$	$N_4(n)$	smallest (i,j)
6	1	1	$(1,2)$
8	3	1	$(2,3)$
10	6	2	$(1,2)$
14	15	2	$(1,5)$
18	28	5	$(1,6)$
22	45	6	$(1,6)$
26	66	7	$(2,3)$
34	120	6	$(5,9)$
62	435	16	$(1,17)$
126	1891	55	$(1,2)$
194	4560	20	$(6,65)$

Table 1: Selected exact counts of four-zero primitive polynomials.

8.2 Finite-pair association in the $n = 2p$ family

For $n = 2p$ and $1 \le i < j \le p-1$, define

$$X_{i,j} = \mathbf{1}_{\mathrm{prim}(f_{2p,i,j})}.$$

For fixed p these are deterministic 0/1 labels on the candidate set

$$I_p = \{(i,j) : 1 \le i < j \le p-1\}, \qquad |I_p| = \binom{p-1}{2} =: N.$$

Put $N_4(2p) = \sum_{i<j} X_{i,j}$ and $\mu = N_4(2p)/N$. To record an association statistic inside this finite list, choose an unordered pair $\{a,b\}$ uniformly from I_p and define

$$\kappa_p = \Pr_{\{a,b\}\subset I_p}(X_a X_b = 1) - \mu^2 = \frac{\binom{N_4(2p)}{2}}{\binom{N}{2}} - \mu^2. \tag{12}$$

This is a finite-population pair-association statistic. It is not the variance of $N_4(2p)$ and we do not use it as a probabilistic variance input.

The exact counts give $\kappa_p < 0$ for all odd primes $5 \le p \le 97$; Table 2 gives selected values. A negative value says that primitive candidates occur in pairs less often than under an independence baseline with the same primitive frequency.

p	N	$N_4(2p)$	μ	κ_p
13	66	7	0.1061	-0.001459
23	231	15	0.0649	-0.000264
31	435	16	0.0368	-0.0000816
41	780	34	0.0436	-0.0000535
61	1770	39	0.0220	-0.0000122
83	3321	66	0.0199	-0.00000587
89	3828	98	0.0256	-0.00000652
97	4560	20	0.00439	-0.000000958

Table 2: Selected finite-pair association statistics for the $n = 2p$ family.

The table records a negative-association signal; the exact counts give $\kappa_p < 0$ for every odd prime $5 \le p \le 97$. It is a diagnostic for the finite candidate set and does not assert a variance bound; in particular, it is not combined with a Chebyshev argument here.

The following conjecture is suggested by the data.

Conjecture 8.1 (Negative finite-pair association)**.** For every odd prime $p \geq 5$,

$$\kappa_p \leq 0.$$

A crude random-polynomial density heuristic suggests that the count $N_4(2p)$ should be of order $p/\log p$: a random monic polynomial of degree $2p$ over $\mathbb{F}_2$ is primitive with probability asymptotic to $\varphi(2^{2p}-1)/(2p\,2^{2p}) \asymp 1/(p\log p)$. This heuristic is not used in the structural results.

8.3 Reproducibility

The computations reported here are reproducible from the following Python scripts available from the author:

- `_analyze_zero_pairs.py`
- `_find_palindromic_primitive.py`
- `_fourzero_n4_two_lines.py`
- `_fourzero_pair_association.py`
- `_extend_fourzero_first_202_250.py`
- `_extend_fourzero_first_252_300.py`

The factor table used by the enumeration is cached in `_factordb_2n_minus_1_cache.json`. The numerical data are available from the author in JSON and Markdown form, including the exact-count table and the finite-pair association table. A general companion repository for related preprints is maintained at the author's GitHub repository; Paper F assets are available from the author.

9 Open problems

The following problems remain open.

Problem 9.1. Prove Conjecture 1.2.

Problem 9.2. Prove the $n = 2p$ subfamily for all odd primes p.

Problem 9.3. Prove Conjecture 8.1, or find a counterexample.

Problem 9.4. Determine the selected factors $\mathcal{S}_p(T) = \gcd(G_p(T), M_p(T))$ without term-by-term coefficient computation.

Problem 9.5. Prove that for $p \geq 13$ the map from four-zero primitive polynomials to τ-minimal polynomials is injective. Computations support this for $p \leq 61$.

10 Discussion

The structural results above show that the four-zero tail-palindromic problem is not an arbitrary coefficient problem: it is governed by a self-paired space, a quotient by a common fixed-point kernel, an explicit half-space functional, and, in the $n = 2p$ case, by a necessary Dickson-factor filter. In the computed range, the finite-pair association statistic for the $n = 2p$ family is negative.

At the same time, the main existence question remains open. A single-candidate character-sum approach cannot suffice: for $n = 2p$ there are about p^2 candidates, whereas generic Weil-type error terms have size about $2^{p/2}$. Thus any analytic proof must exploit cancellation across the entire four-zero family, for example through a second-moment or large-sieve estimate for a probabilistic model of the primitive-event family.

A Auxiliary combinatorial and arithmetic descriptions

This appendix records two auxiliary descriptions of the four-zero candidate set. They are not used in the Dickson reduction or in the existence arguments, but they clarify the combinatorial and arithmetic structure of the search domain.

Proposition A.1 (Insertion poset)**.** *Let $k \geq 3$ and $n = 2k$. Four-zero tail-palindromic candidates of length n are in bijection with triples*

$$(x, y, z) \in \mathbb{Z}^3_{\geq 0}, \qquad x + y + z = k - 3.$$

The corresponding half-code is

$$H_{x,y,z} = 1^{x+1}0\,1^y0\,1^z.$$

The number of candidates is therefore

$$\binom{k-1}{2}.$$

Moreover, symmetric insertion corresponds to the three directed moves

$$(x, y, z) \to (x + 1, y, z), \quad (x, y, z) \to (x, y + 1, z), \quad (x, y, z) \to (x, y, z + 1).$$

Thus the candidate set is the $(k-3)$-th layer of a three-dimensional Pascal poset, and the number of paths from $(0, 0, 0)$ to (x, y, z) is the multinomial coefficient

$$\binom{k-3}{x, y, z}.$$

Proof. A tail-palindromic code of length $2k$ is determined by its first half of length k. The four zeros occur in two symmetric pairs. In the first half, write the two zeros with gaps x, y, z, so that the half-code is $1^{x+1}0\,1^y0\,1^z$. Then

$$(x + 1) + 1 + y + 1 + z = k,$$

hence $x + y + z = k - 3$. Conversely, every such triple determines a four-zero tail-palindromic code by reflection.

Adding one to x, y, or z is exactly the insertion of a pair of 1's at one of the three gaps while preserving palindrome. Therefore the directed graph is the three-dimensional Pascal poset, and the path count is the standard multinomial coefficient. □

Proposition A.2 (Symmetric-weight gcd)**.** *Let $n = 2k$ and let $(i, n-1-i)$ be a symmetric zero pair of a four-zero candidate. Define*

$$A_{n,i} = 2^i + 2^{n-1-i}.$$

Then

$$\boxed{\gcd(A_{n,i}, 2^n - 1) = 2^{\gcd(k,2i+1)} + 1.}$$

The same formula holds with i replaced by the other zero position j.

Proof. Write $d = n - 1 - 2i$. Since n is even, d is odd, and

$$A_{n,i} = 2^i(1 + 2^d).$$

Because $\gcd(2^i, 2^n - 1) = 1$,

$$\gcd(A_{n,i}, 2^n - 1) = \gcd(2^d + 1, 2^n - 1).$$

Let $e = \gcd(n, d) = \gcd(n, 2i + 1)$. Since $n = 2k$ and $2i + 1$ is odd, this equals $\gcd(k, 2i + 1)$. The inclusion $2^e + 1 \mid \gcd(2^d + 1, 2^n - 1)$ is immediate from the two divisibilities.

Conversely, let ℓ be a prime divisor of $\gcd(2^d + 1, 2^n - 1)$, and let $h = \mathrm{ord}_\ell(2)$. Then $h \mid n$ and $h \mid 2d$ but $h \nmid d$. Since d is odd, $h = 2f$ with $f \mid d$ and $f \mid n$. Hence $f \mid e$, and because e/f is odd, $\ell \mid 2^e + 1$. Therefore the two gcds are equal. □

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